

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019,
ICRTAC 2019

BLE Bluetooth Beacon based Solution to Monitor Egress of Alzheimer's Disease Sufferers from Indoors

D.Surendran^a, M.Rohinia^a

^a*Sri Krishna College of Engineering and Technology, Coimbatore, 641008, India.*

Abstract

With the ageing people community, Alzheimer's disease (AD) is the most common disease in prevalence. It affects the memory of the affected individual by causing neuron degeneration. Such cognitive disability leads people to roam around places since they are ignorant about location. They may go afar from the safe area to traffic prone areas which cause serious, life threatening issues to them. Proposed study introduces a technology that involves a wearable device for Alzheimer disease affected elderly patients and listening device given to care takers or doctors. This system detects and notifies caregivers about the departure of people to risk locations from their residence zone. This system uses Bluetooth Low energy (BLE) communication technology that utilizes transmitter and receiver fixed to elderly individual as wearable device and in residence respectively. Based on the signal strength between the transmitter and receiver, the distance estimation between the elderly and the receiver is calculated with the help of a Raspberry Pi(Rpi) and an alert notification is sent to doctors/caretakers when the distance exceeds safety limit.

© 2019 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019.

Keywords: Alzheimer's disease, BLE (Bluetooth Low Energy), Wearable Bluetooth beacons, Raspberry Pi.

1. Introduction

Alzheimer's disease is age related pathology that causes several symptoms, consequences to both patients and care givers. Onset of symptoms occurs several years before the onset of disease. They have difficulty in planning, remembering and communicating. Their mood swings varies often and cause high anxiety and fear. They often do not want to participate in any social meetings. They have difficulty in recognizing visual images, crossing roads, identifying routes and travelling in vehicles. They need extra care-givers all the time and cannot decide anything on their own[1]. In the global population, people getting older is growing with the ratio from 28% in 2009 to 38% in 2050. This phenomenon has been referred to as increase of global world population ratio of getting old. Several reasons for this phenomenon includes reduced degrees of person's mortality, decrease in overall transportation costs. There are 800,000 individuals suffering AD in the UK and the data is expected to increase more than one million by 2030. 60–80% elderly population with dementia symptoms are of the Alzheimer's disease (AD) (Alzheimer's Association, 2018). Hence this is called as old associated Alzheimer's pathology that prevails in majority of population and proposing solution and aid for those inhabitants is of great necessity. The early symptoms of the AD pathology include loss of navigation intention skills. People with Mild Cognitive Impairment(MCI) will have the mild symptoms and can remember most familiar environments still has difficulty in learning to new environment learning new environments becomes especially difficult. It is regrettable that 80% of people with AD pathology dwell in AD assisted living environments and homes with the help of care takers. For ease of such environments, the proposed design aids such elderly inhabitants and their residence with egress detection and monitoring. Further implementations in design enables impaired abilities and reduce egress amongst AD affected community.

*Corresponding author. Tel.: +919080717071.

E-mail address: surendran@skcet.ac.in

1877-0509 © 2019 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019.

10.1016/j.procs.2020.01.053

This would also improve quality of life and well-being, allow for the highest possible degree of independence to be maintained, reduce the work load of the carers and ease the transition of moving into care-homes [10]. At present there are several technological solutions for assisting affected individuals. Such technological systems can also be implemented into the aged people community that are integrated with several automated software and applications. Such systems designed as wearable devices have been a promising approach to reduce the overhead of care to people.

The problems due to disease pathology can lead to several number of issues such as exit of patients from indoors, get into heavy traffic areas and have life threatening problems [5].

2. Related work and literature review

There are several ideas that distinguish departure and area of people exist, be that as it may, these have a few inadequacies making them inadmissible for utilize given the limitations of the implementation.

Bakshi, G.S.[1] et al. have delivered an application based solution to screen the area of people with Alzheimer's pathology and Autism Spectrum Disorder. This arrangement precisely finds Bluetooth Low Energy (BLE) based gadgets, for example, those put on an in danger individual, inside a domain utilizing various Bluetooth audience members. These Bluetooth audience members screen the quality from Bluetooth LE gadgets. If an individual is in a risky territory the care takers are alarmed. Every application inside this system is conveyed to end user gadgets, for example, telephones and tablets, possessed by volunteers. Every application in that system filters their region for the identifiers of Bluetooth LE signals that have detailed on navigation. On the chance that an application occasion watches such implementation, at that point the area of that smart phone is resolved however GPS and handed-off to a focal administration. This arrangement has indicated guarantee that as it may, it has various shortages which make it inconsistent with the prerequisites. This arrangement has inadequacies for the ideal use case identified with the number of elderly people and care takers demonstrating required for task. In spite of the fact that this could conceivably be utilized to identify Alzheimer's sufferers, it was not assessed to be utilized in this limit. O'Malley et al.[2] contrived an answer for indoor tracking of people inside a situation. This requires an individual to be observed to convey an advanced system of individual which contains an application which uses GPS to track their orientation. This area information is then handed-off to a handling administration. This area is accordingly looked at against guardian indicated "zones" to decide whether an in danger individual is in risky places, for example, the outside of a consideration home. The sign quality information from these audience members is handled by a trilateration calculation which is expanded with Kalman channels. The yield of this preparing is exceedingly precise area of Bluetooth LE gadgets. This arrangement could be adjusted to decide area of an individual and age of alarms of when they enter a dangerous zone. Prominently, it has a dependence on outside GPS signals, has a short battery life and depends on the gadget to be conveyed by an in danger individual to have a web communication. Lin et al. [6] proposed an answer for track people utilizing Bluetooth LE based reference points. In this arrangement people of intrigue were issued with Bluetooth LE based labels, commonly to be incorporated into their garments or individual effects. Radio Frequency identification (RFID) based following arrangements, for example, [9,10], might be adjusted to follow clients area to decide their essence with respect to risky zones. These arrangements influence trilateration of RFID gadgets through numerous audience stations. Detached RFID labels have a favorable position where they don't require a nearby power source. Notwithstanding, much like other trilateration-based methodologies, this arrangement requires broad foundation and complex demonstrating required for solid activity. At whatever point their meandering from a private domain is seen via parental figures, the one of a kind identifier for the Bluetooth LE signal is distributed to a publicly supported system of utilizations. Thusly, this focal administration will hand-off the area to concerned gatherings, for example, guardians and relatives. This arrangement has various inadequacies. Remarkably it depends on guardians to see the nonappearance of an individual and distribute their Bluetooth reference point identifier. Additionally, it depends upon a group based establishment to exist and have forehead individuals inside the region of a missing person. A practically identical arrangement has been created.[6,7].

3. Proposed work

The best way to address these issues is to devise a system that includes a wearable device worn by AD elderly people which tracks their navigation indoors. It is secure beacon technology that monitor the depart of affected people from their safe indoor area. The proposed study provides the population of the AD patients with wearable Bluetooth Low Energy beacons. Bluetooth beacons are devices which broadcast information and data such as temperature, their location identity and truly the presence of individuals. This enables to incorporate a unique identifier for each beacon. These beacons are listened by Bluetooth receivers that are present in entrance and exit of indoors. These receivers display unique identifier code of BLE beacons which gives the location of the individual and the receiver signal. The receiver units comprises of electromagnetic unit that attenuate the false positive signals occurring when other neighbors roam near the AD individuals. These receiver signal data are given to the Pushwoosh RSSI that implements Sensor information for android. This individual beacon wise ID, Receiver signal are processed by alerting systems that issue alert top care takers when receiver signal value reaches limit.

Alerting is issued by Sensor monitoring APK Pushwoosh mobile notification application. This application is installed with the mobile phones of care takers who receive notification about the location details of beacon, receiver signal so that they can monitor the location of AD individuals and also monitor their surrounding whether they are alone or loitered by neighboring crowd as given in Figure 1.

3.1. Bluetooth beacons Based Motion Detection

Bluetooth beacons are small and less expensive device that are used for indoor positioning, monitoring of departure of individual from one location and detects their presence in particular place. With the receiver systems that are Bluetooth listeners the data received can be invoked by the alert management applications, It can be combined to the personal gadgets as mobile applications to the care takers or doctors. For proper listening of beacons, location is important feature and must be properly placed. In the present study, beacons are mounted in wearable devices like wrist watch. Precise locations are column partitions of device that they need to be placed in order that there's clear observation between the caretakers mobile phone and the wearable beacon location with no interference in between. It can also be checked for double-sided tape to vicinity the beacons but make certain that they are consistent in a set location and attached properly. Beacons must be place at the peak of 2 to 3 meters out of arms to keep away from strange observers tampering intentionally. Reports are generated to the care takers and doctors about the details of location the patients, where the elderly persons have been, the time and how many times they have departed from indoor .

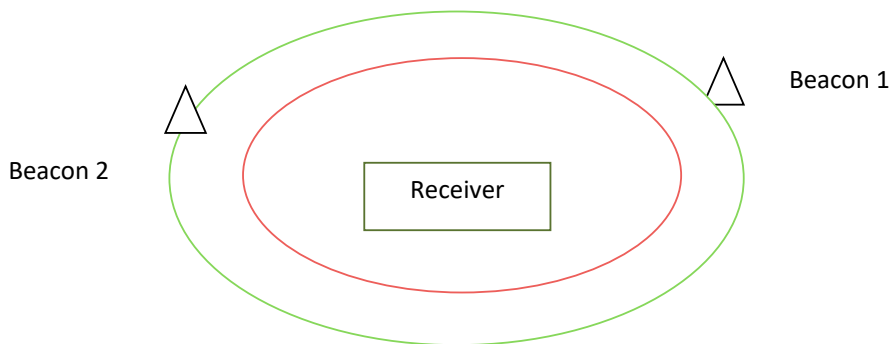


Fig.1. Listener values detected by receiver from two beacons (embedded in wearable device).

This proof of concept is presented in Figure 2. that notifies when elderly person egress from one room to the other room with significant interferences with neighbors that can be given as receiver signals to the listeners.

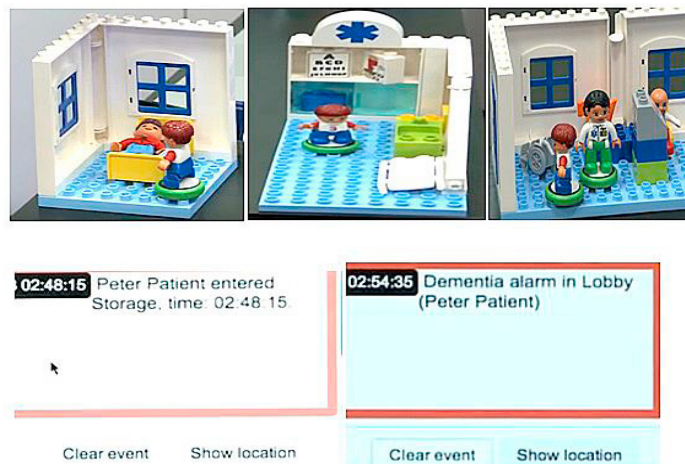


Fig.2. A proof of concept listener to evaluate motion detection with elderly inhabitants.

4. System Architecture

The design of the proposed system is the motivation to create monitoring and alerting system. It provides family members, doctors and care-takers to get updated on the present location of elderly people indoors and their wandering details. The cost of final product is made feasible by simplifying the modules instead complex codes that involves many computations and storage. The proposed study consists of three modules: beacon broadcast system, proximity estimation, alerting program and pushwoosh alert. The transmitter is beacon BLE wearable watch that is carried by the AD affected elderly person and is paired with the receiver. The receiver can also be a Bluetooth beacon, but in the proposed system a smartphone implemented. The listener to the beacon is a commercial Bluetooth Dongle fixed in object or furniture inside the house. Both the relay system and Bluetooth listener are paired together that attenuates the interference from neighbours loitering nearby the relay system. The Bluetooth listener dongle embedded in the house is designed and implemented with Raspberry Pi via USB and given suitable power supply. Raspberry Pi is programmed with proximity estimation module that accepts input parameters from the listener. Listener receives three parameters. The Bluetooth application software is implemented and is invoked by a shell program that will be running once the transmitter, receiver and smartphone are set. Using that program the receiver signal strength value of the connected bluetooth enabled beacon device can be retrieved.

The Pushwoosh mobile notification system consists of two modules. First module is alerting program alertPY(), that supports back ends servers for pushing alert messages to mobile devices. The second module is android application that will be installed on the caretakers, doctors smartphone. The application will receive messages sent from the Raspberry Pi through the alertPY() program. For multiple notification about elderly activity additional module is written in the application.

The Proximity Estimation and Push-Alert systems are handled using a shell program. The program is implemented with monitoring of receiver signal strength and beacon id for proximity estimation. The proposed system architecture is given in Figure 3.

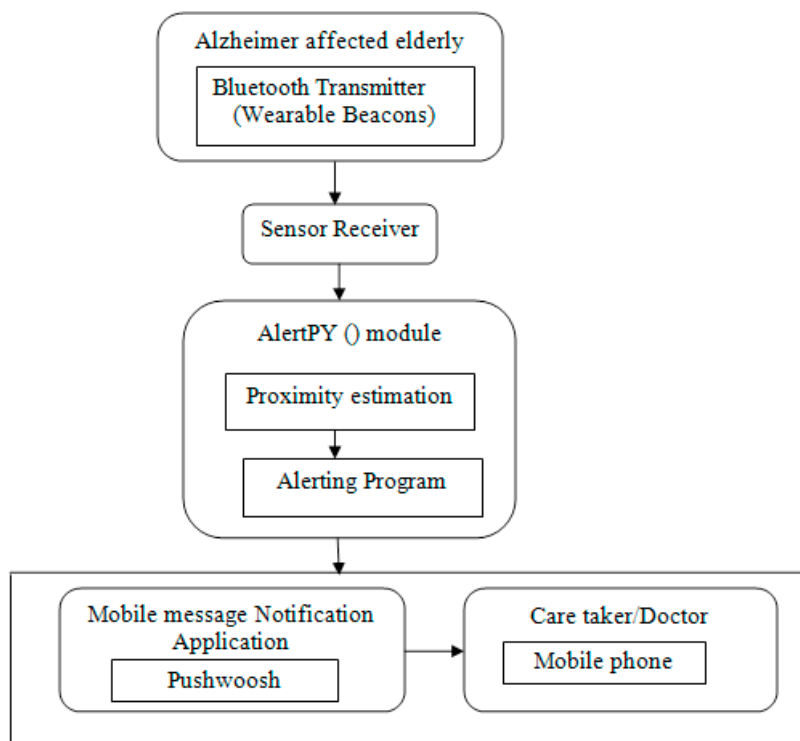


Fig.3. BLE Bluetooth Beacon based Egress Monitoring Architecture

4.1. Hardware Components

The hardware components of this solution are classified as wearable beacons, listeners and smart phones. Bluetooth beacon power includes connectivity of about 70 meters far that depends on the power saved with the wearable device

embedded with beacons. The device does not support back communication channel since unidirectional. The device is wearable Bluetooth low energy with Ultra-low power consumption chipset nRF52 series that supports 5url broadcasting with android & IOS SDK provided. Its size(H,W,L) is 244,38,10mm, CR2032 Battery model, Battery capacity of 240mAh. Wearable bluetooth beacons transmit their unique id and transmission power such as signal strength, temperature or the other values programmed by the user. When the smartphone gets connected to beacon device, identification depends on the mobile application installed. It receives notification or application based id from beacon embedded device. The distance between beacon and the mobile phone is calculated using signal strength. When the mobile application is initiated, it notifies all the information about connected multiple beacons within the limited range and their positions are triangulated. Bluetooth beacon constitutes Bluetooth chipset that manages indoor positioning, unidirectional property and accurately provides proximity information. It works with Bluetooth Low Energy technology that is Bluetooth 5 with all specification upto Bluetooth 4 embedded with it. Recently, it is of great importance for numerous positioning applications in markets mainly because of its principle of low energy consumption. The transmission range is higher that allows whole home implementation as wireless smart home Rather than WIFI, Zigbee that uses much power it is much compatible for positioning with low power consumption and also supports mesh networking with smart phone. It has high broadcasting message capacity and the beacon frame size ranges upto 255 bytes in single frame.



Fig.4. a)Wearable Bluetooth beacons embedded in the wrist watch (b) Wearable Bluetooth beacons embedded in the Lanyard

In addition to wearable Bluetooth beaconst transmitters, receivers are needed. The receivers are smart phones that are implemented with android application of listening code that are invoked by shell program. The Android smart phones can be a Moto G/Nexus 5. Its processing board constitutes a Raspberry Pi. Suitable for receiving data frames and are tested with beacon wearable device for proximity. It is also tested for interference with shielding of transmitters. The data from these receiver components are given as input to alertPY() module and processed by the alert program and emergency condition is identified.

4.2. Alert Management

The alert program identifies whether the AD affected elderly has departed from their residence and notifies the alert management application installed. This alert program listens these notification with Pushwoosh mobile application as interface .On receiving the notification, the application will alert doctors, care-takers with message or alarm that can be programmed by the user. Each smart phone can be connected to the multiple beacon devices that can be implemented in elderly people residence that notifies with beacon id and receiver signal and identifies whether motion is detected. Alert management module supports multiple beacon devices worn by elderly inhabitants in home and each are notified with their individual ID. alertPY() module invokes several module depending on each beacon identity and receives the distance calculated between gadgets and the elderly people. Input variables given to this function are recursively varies for each id i.e for each individuals. Alert notification are also issues as per the design of output from mobile applications.

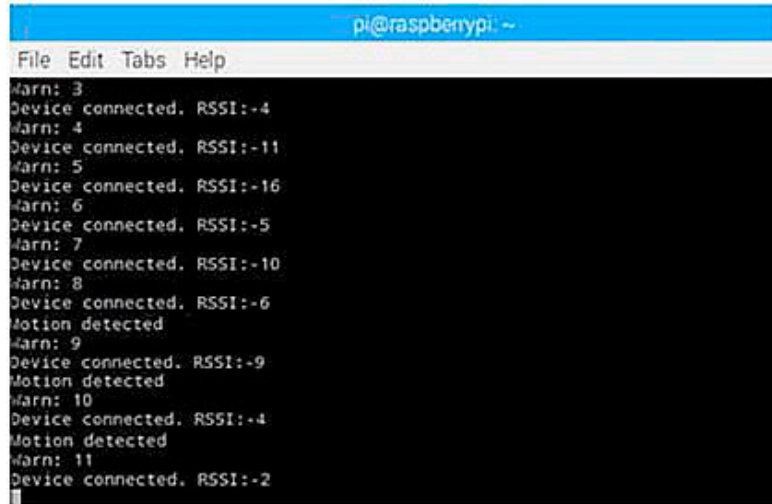


Fig.5.Receiver Signal values listened from transmitter.

5. Evaluation

The framework was set up and arranged. The Smartphone is associated with the Raspberry Pi utilizing the Bluetooth Dongle. The shell program was the startup code, invoked and connections were checked. The program would continue to associate with the gadget and survey for receiver signal perusing at customary 0.1 second interims. The receiver signal readings could be characterized into 3 positive, zero and negative values. Positive readings demonstrated that the transmitter was exceptionally near the recipient (around 110 centimeters). In the event that the receiver signal value was zero, it showed that the transmitter was in the general region of the elderly individual (around a foot or two). When the values were negative, the transmitter was at a separation from the collector. Furthermore, the push notification messages were sent to the Smartphone. Receiver signal readings are recovered from the transmitter. And hence the push notifications were sent when the receiver values changed to negative for the elderly individuals' navigation. Receiver signal readings were retrieved from the transmitter. The receiver signal value changed to positive when the transmitter is closer to the receiver and it decreases to zero when moved away from the receiver. It is noted that the receiver signal values changes to the negative value when the distance from the elderly person and the receiver increases. This prevented us from calculating the distance from the transmitter and receiver accurately. But the values of receiver signal were consistently negative when the transmitter was at a distance from the receiver. The other issue that was observed was that if there was a line of sight between the transmitter and receiver, the RSS value would be constant at zero until the transmitter was moved away more than 10 feet. An ingenious solution helped solve this issue by placing the transmitter around the neck of the wearer. When placed in such a position, the line of sight is broken every time the wearer moves away from the receiver.

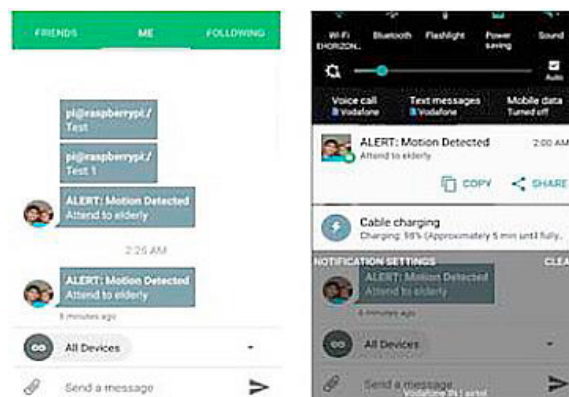


Fig.6. Alert notifications from Pushwoosh interface successfully received on the caretaker's smartphone.

6. Conclusion and Future Work

This study has used Bluetooth LE principle that can monitor exit/departure of AD affected individuals of a residential environment. This solution addressed the disadvantages in present egress monitoring approaches. This proposed study that monitor egress of aged community with suitable alerts to care-takers, doctors particularly when aged community are with the pathology of age related illness such as Dementia, AD. In these cases the proposed study may be lifesaving. The system would constantly listens the receiver strength values from the transmitter and send alert notification when the receiver strength values decreases below the threshold value for a fixed duration. Cost of the system was kept to a minimum and has been identified to be comparatively cheaper than the currently available systems. In the future, the system can be further improved by choosing best learning algorithms that accurately determine the receiver signal value of the wearable device transmitter. The movement of the AD affected elderly individual can be tracked throughout the set time period and analyzed for patterns in the time of day which could be improved with the help of learning algorithms. Other advantages may include reducing of interference of loiters by shielding for better identification of departure zone or learning algorithms that provide departure locations to be redefined though modification of software. The use of other positioning technologies such as active RFID tags and Ultra-wideband band markers may be studied in future that compares each technology and provides the most compatible implementation.

References

1. Bakshi, G.S.; Chen, Z. Baby boom, population aging, and capital markets. *J. Bus.* 1994, 67, 165–202.
2. O'Malley, M.; Innes, A.; Wiener, J.M. Decreasing spatial disorientation in care-home settings: How psychology can guide the development of dementia friendly design guidelines. *Dementia* 2017, 16, 315–328.
3. Zingmark, K. Promoting a good life among people with Alzheimer's disease. *J. Adv. Nurs.* 2002, 38, 50–58, doi:10.1046/j.1365-2648.2002.02145.x.
4. Peng, L.-M.; Chiu, Y.-C.; Liang, J.; Chang, T.H. Risky wandering behaviors of persons with dementia predict family caregivers' health outcomes. *Aging Ment. Health* 2017, 1–8, doi:10.1080/13607863.2017.1387764.
5. Reisberg, B.; Borenstein, J.; Salob, S.P.; Ferris, S.H. Behavioral symptoms in Alzheimer's disease: Phenomenology and treatment. *J. Clin. Psychiatry* 1987, 48, 9–15.
6. Lin, Q.; Zhang, D.; Chen, L.; Ni, H.; Zhou, X. Managing elders' wandering behavior using sensors-based solutions: A survey. *Int. J. Gerontol.* 2014, 8, 49–55, doi:10.1016/j.ijge.2013.08.007.
7. Lin, Y.-J.; Chen, H.-S.; Su, M.-J. A cloud based Bluetooth Low Energy tracking system for dementia patients. In Proceedings of the 2015 Eighth International Conference on Mobile Computing and Ubiquitous Networking (ICMU), Hakodate City, Japan, 20–22 January 2015; pp. 88–89.
8. Mhamdi, J.; Abkari, S. El Contriving an RFID system for Alzheimer patients tracking. In Proceedings of the 2015 Third International Workshop on RFID And Adaptive Wireless Sensor Networks (RAWSN), Agadir, Morocco, 13–15 May 2015; pp. 23–28.
9. López-de-Ipiña, D.; Díaz-de-Sarralde, I.; Zubia, J. An Ambient Assisted Living Platform Integrating RFID Data-on-Tag Care Annotations and Twitter. *J. Univ. Comput. Sci.* 2010, 16, 1521–1538.
10. Garcia, C.G.; Meana-Llorián, D.; G-Bustelo, B.C.P.; Lovelle, J.M.C.; Garcia-Fernandez, N. Midgar: Detection of people through computer vision in the Internet of Things scenarios to improve the security in Smart Cities, Smart Towns, and Smart Homes. *Futur. Gener. Comput. Syst.* 2017, 76, 301–313.